MVP Feedback & Architectural Review

Project: ProdLog / Diligence (Product Decision Funnel)

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1. Executive Summary

The **Diligence** / **ProdLog** architectural design and product definition documents present an exceptionally well-disciplined framework for high-leverage AI integration. Unlike conventional AI tools that fall into the trap of "lazy ghostwriting" (i.e., generating low-quality content, writing PRDs from scratch, or fabricating insights), this project leverages large language models (LLMs) as a **forcing function for human rigor**.

By establishing an "anti-generative" paradigm, the tool positions itself as an automated product mentor that maps, tracks, and scores the qualitative journey of product judgment. The architecture relies on a strict Bring-Your-Own-Key (BYOK) paradigm, ensuring a genuine `$0 running cost` at any scale.

> **Core Thesis Approval:** Traditional PM infrastructure focuses exclusively on tracking execution (Jira) or indexing raw requirements (Productboard, Notion). Tracking the *underlying rationale, evidence gaps, and explicit validation overrides* creates an entirely new category: an immutable audit trail for product judgment.

2. Strategic Wins (What is Highly Effective)

2.1 The "Logged Overrides" Principle

Enforcing rigorous gate compliance is historically why stage-gate processes fail in agile cultures—they introduce friction. By allowing the PM to explicitly **skip any gate** provided they supply a one-line justification, you preserve development momentum while seamlessly constructing an audit trail of compromised rigor. Visibility, rather than mechanical enforcement, is the correct organizational lever.

2.2 Radical Scope Discipline

Limiting the V1 archetype strictly to your own B2B SaaS environment (e.g., managing specific platform lines like GP and LP dashboards) eliminates the risk of over-generalized, sycophantic prompt phrasing. Tuning context specifically to a narrow business line yields a dramatic increase in AI feedback quality.

2.3 Gate 9 Post-Launch Loop Closure

Product teams are notoriously susceptible to shipping a feature and immediately racing to the next initiative without validating real-world performance against initial hypotheses. Isolating "shipped but unreviewed" features on the primary dashboard dashboard addresses a profound cultural blind spot in product management.

3. Critical Architectural & Data Model Gaps

While the execution blueprint is clean, there are structural mismatches between the planned AI behavior and the underlying data schema that must be resolved prior to Phase 1 development.

3.1 The Multi-Turn Chat Schema Gap

> **The Issue:** In Section 5.2 of the Implementation Plan, the spec states: *"If the user's response does not meet the evidence bar, ask AT MOST ONE probing follow-up, then stop."*

However, the original proposed JSON state model accommodates only a single `questionAsked` and a single `response` per gate:

```
// Original Vulnerable Schema
evidence: [
  {
    gateOrder: number,
    questionAsked: string,
    response: string,
    status: "provided" | "skipped",
    skipReason: string | null,
    timestamp: ISOstring
  }
]
```

If the AI evaluates the initial user entry, finds it deficient, and prompts a secondary probing follow-up question, there is no allocated state schema to record that secondary exchange. Overwriting the fields corrupts the historical ledger; appending text manually creates a parsing nightmare.

The Fix: Restructure the gate storage to contain an explicit conversation array, keeping the historical interaction distinct from the final synthesized evidence ledger.

```
// Recommended Robust Schema
evidence: [
  {
    gateOrder: number,
    status: "provided" | "skipped",
    skipReason: string | null,
    timestamp: ISOstring,
    conversation: [
      { role: "assistant", text: "Personalized core question..." },
      { role: "user", text: "Initial evidence entry..." },
      { role: "assistant", text: "Probing follow-up question..." },
      { role: "user", text: "Elaborative secondary response..." }
    ],
    finalEvidenceSummary: string // Extracted or direct answer for the record
  }
]
```

3.2 Anthropic CORS Realities & Proxy Requirements

While Anthropic's API endpoints technically support cross-origin resource sharing (CORS) for direct browser interaction under certain configurations, building production-ready error handling client-side can be fragile. Expired keys, network connection transitions, or sudden schema variations can break direct browser calls.

Mitigation: Plan to deploy a lightweight, stateless Cloudflare Worker early in Phase 2. This worker will merely forward requests while intercepting headers. This guarantees full compatibility with strict browser sandboxes, maintains the zero-cost architecture within Cloudflare's free tier limits, and protects your main app logic from direct API variation.

3.3 LocalStorage Ceiling Thresholds

Standard browser `localStorage` enforces a strict volume ceiling of approximately 5MB per origin. If you begin embedding extensive user interview transcripts, customer support tickets, or database dumps within the free-text fields across multiple decisions, you will exhaust this boundary faster than anticipated.

Mitigation: The JSON export/import data mechanism scheduled for Phase 8 should be promoted directly to **Phase 1**. This acts as an immediate insurance policy against browser state clearing and gives you a mechanism to archive historical decisions out of active localStorage memory if necessary.

4. UX & Concept Refinements

4.1 De-gamifying the Diligence Indicator

The PRD raises an open question regarding whether the **Diligence Score** should be qualitative or a singular quantitative metric (e.g., 78%).

Recommendation: Use Qualitative Archetypes. Assigning a numerical value forces standard PM optimization behavior: users will instinctively dump low-value text or fluff into inputs simply to game the engine up to a high score. Instead, assign prominent, descriptive archetypes based on *where* the skips occur:

Diligence Persona	Skipped Trigger Pattern	UX Backlog Tag & Impact
High Risk Velocity	Skipped Gate 2 (Problem Validation) and/or Gate 5 (Solution Validation)	**UNVALIDATED** Warns that building has commenced without external market signals.
Impact Blind	Skipped Gate 4 (Opportunity Sizing) or Gate 7 (Trade-offs)	**SIZING MISSING** Highlights that the engineering effort is decoupled from clear ROI.
Rigorous Build	All validation, alignment, and scoping gates fully executed.	**FULLY EVIDENCED** Signals high internal confidence and structural clarity.

5. Phase Alignment & Tactical Execution

To maximize velocity, ensure your Phase 1 testing is highly adversarial. When standing up the application prior to wiring up the Anthropic API, manually simulate erratic behavior:

- Create initiatives and skip 4 gates in rapid succession to ensure your qualitative tags transition accurately on the backlog dashboard view.
- Validate that the manual entry of long-form markdown text functions smoothly in the table rows without destabilizing layout alignment.
- Verify that your date sorting isolates items stuck at Gate 9 (Post-Launch Review) dynamically.

By establishing this structural baseline first, your AI layer integration in Phase 2 will be a straightforward exercise in prompt engineering rather than a complex infrastructural overhaul.